Chapter 4. Rings and Fields

Rings, integral domains, Fermat and Euler, fields of quotients, polynomial rings, factorization, quaternions, ordered rings

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4.1 Rings and fields (§18)

Motivation. $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, polynomials and matrices carry two operations linked by the distributive law. Rings are the common abstraction; fields are the rings in which division is possible, the setting of linear algebra and of solving polynomial equations.

Definition (Ring)

A **ring** $\langle R, +, \cdot \rangle$ is a set with two binary operations such that
$\mathcal{R}_1$: $\langle R, + \rangle$ is an abelian group;
$\mathcal{R}_2$: multiplication is associative;
$\mathcal{R}_3$: both distributive laws $a(b+c) = ab + ac$, $(a+b)c = ac + bc$ hold.
R is **commutative** if multiplication is; a **unity** is a multiplicative identity $1 \neq 0$; a **unit** is an element with a multiplicative inverse. A **division ring** (skew field) is a ring with unity in which every nonzero element is a unit; a **field** is a commutative division ring.

Theorem 4.1 (18.8)

In a ring, (1) $0a = a0 = 0$;
(2) $a(-b) = (-a)b = -(ab)$;
(3) $(-a)(-b) = ab$.

Proof. (1) $a0 + a0 = a(0+0) = a0 = a0 + 0$; cancel in the additive group. Similarly $0a = 0$.

(2) $ab + a(-b) = a(b + (-b)) = a0 = 0$, so $a(-b) = -(ab)$; likewise $(-a)b = -(ab)$.

(3) $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$. ■

A **ring homomorphism** $\phi : R \to R'$ satisfies $\phi(a+b) = \phi(a) + \phi(b)$ and $\phi(ab) = \phi(a)\phi(b)$; an **isomorphism** is a bijective homomorphism. Examples: $\mathbb{Z}_n$; $\mathbb{Z} \times \mathbb{Z}$; $M_n(R)$; the ring of functions $\mathbb{R} \to \mathbb{R}$; the reduction map $\mathbb{Z} \to \mathbb{Z}_n$. By the Chinese remainder argument, $\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n$ as rings when $\gcd(m,n) = 1$ ($k \mapsto (k \bmod m, k \bmod n)$ is a ring homomorphism, and an additive isomorphism by Theorem 2.14).

4.2 Integral domains (§19)

Motivation. In $\mathbb{Z}_6$ we have $2 \cdot 3 = 0$, so the familiar rule "$ab = 0$ implies $a = 0$ or $b = 0$" fails, and with it cancellation and the usual way of solving equations. Integral domains are exactly the rings where cancellation survives; every finite one is a field, and the characteristic records how the integer multiples of 1 behave.

Nonzero a, b with $ab = 0$ are **divisors of** 0.

Theorem 4.2 (19.3)

The divisors of 0 in $\mathbb{Z}_n$ are exactly the nonzero elements not relatively prime to n.

Proof. If $m \neq 0$ and $d = \gcd(m,n) \neq 1$, then $m \cdot (n/d) = (m/d)n = 0$ in $\mathbb{Z}_n$ with $n/d \neq 0$. If $\gcd(m,n) = 1$ and $ms = 0$ in $\mathbb{Z}_n$, then $n \mid ms$, so $n \mid s$ and $s = 0$. ■

Corollary 4.3 (19.4)

If p is prime, $\mathbb{Z}_p$ has no divisors of 0.

Theorem 4.4 (19.5)

The cancellation laws ($ab = ac$, $a \neq 0 \Rightarrow b = c$, and similarly on the right) hold in R iff R has no divisors of 0.

Proof. If cancellation holds and $ab = 0 = a0$ with $a \neq 0$, then $b = 0$. Conversely, if there are no divisors of 0 and $ab = ac$, $a \neq 0$, then $a(b - c) = 0$, so $b - c = 0$. ■

An **integral domain** is a commutative ring with unity and no divisors of 0.

Theorem 4.5 (19.9)

Every field is an integral domain.

Proof. If $ab = 0$ and $a \neq 0$, then $b = a^{-1}ab = 0$. ■

Theorem 4.6 (19.11)

Every finite integral domain is a field.

Proof. For $a \neq 0$, $x \mapsto ax$ is one-to-one (cancellation), hence onto the finite set D; so $ax = 1$ for some x. ■

Corollary 4.7 (19.12)

If p is prime, $\mathbb{Z}_p$ is a field.

The **characteristic** of R is the least $n \in \mathbb{Z}^+$ with $n \cdot a = 0$ for all $a \in R$, or 0 if there is none.

Theorem 4.8 (19.15)

Let R have unity. If $n \cdot 1 \neq 0$ for all $n \in \mathbb{Z}^+$, R has characteristic 0; otherwise the least n with $n \cdot 1 = 0$ is the characteristic.

Proof. If $n \cdot 1 = 0$, then $n \cdot a = (n \cdot 1)a = 0$ for all a; conversely $n \cdot a = 0$ for all a includes $a = 1$. ■

The characteristic of an integral domain is 0 or prime: if $n = rs$ with $1 < r, s < n$, then $(r \cdot 1)(s \cdot 1) = 0$.

4.3 Fermat's and Euler's theorems (§20)

Motivation. Lagrange's theorem, applied to the group of units of $\mathbb{Z}_n$, gives classical number theory almost for free: $a^{p-1} \equiv 1 \pmod{p}$ and $a^{\varphi(n)} \equiv 1 \pmod{n}$. The section also settles when the linear congruence $ax \equiv b \pmod{m}$ is solvable.

Theorem 4.9 (20.1, little theorem of Fermat)

If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.

Proof. $\mathbb{Z}_p^*$ is a group of order $p-1$ (Corollary 4.7), so the order of $\bar{a}$ divides $p-1$ (Theorem 2.11). ■

Corollary 4.10 (20.2)

$a^p \equiv a \pmod{p}$ for every integer a and prime p.

Proof. Multiply Fermat by a; if $p \mid a$ both sides are 0. ■

Theorem 4.11 (20.6)

The nonzero elements of $\mathbb{Z}_n$ that are not divisors of 0 form a group G_n under multiplication modulo n.

Proof. They are the m with $\gcd(m,n)=1$ (Theorem 4.2); products of such are such; $1 \in G_n$; and for $a \in G_n$, $x \mapsto ax$ is one-to-one on the finite set G_n (cancellation), hence onto, so $ax=1$ for some $x \in G_n$. ■

$\phi(n) = |G_n|$ is **Euler's phi-function**.

Theorem 4.12 (20.8, Euler)

If $\gcd(a,n)=1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$.

Proof. Lagrange in G_n. ■

Theorem 4.13 (20.10)

If $a \in \mathbb{Z}_m$ is relatively prime to m, then $ax=b$ has a unique solution in $\mathbb{Z}_m$ for each b.

Proof. a is a unit of $\mathbb{Z}_m$; $x = a^{-1}b$, unique by cancellation. ■

Corollary 4.14 (20.11)

If $\gcd(a,m)=1$, the solutions of $ax \equiv b \pmod{m}$ form exactly one residue class modulo m.

Theorem 4.15 (20.12)

Let $a, b \in \mathbb{Z}_m$ and $d = \gcd(a,m)$. Then $ax=b$ has a solution in $\mathbb{Z}_m$ iff $d \mid b$, and then it has exactly d solutions.

Proof. If $ax=b$ in $\mathbb{Z}_m$, then $ax-b=qm$ in $\mathbb{Z}$, so $d \mid b$. Conversely, if $d \mid b$, write $a=da_1$, $b=db_1$, $m=dm_1$. In $\mathbb{Z}$, $ax-b=qm$ iff $a_1x-b_1=qm_1$, so the solutions are the integers $x \equiv x_0 \pmod{m_1}$, where x_0 is the unique solution of $a_1x \equiv b_1 \pmod{m_1}$ (Theorem 4.13, as $\gcd(a_1,m_1)=1$). Modulo m these are $x_0, x_0+m_1, \ldots, x_0+(d-1)m_1$, which are d distinct classes. ■

Corollary 4.16 (20.13)

$ax \equiv b \pmod{m}$ is solvable iff $d=\gcd(a,m)$ divides b, and then the solutions form exactly d residue classes modulo m.

Example 4.1

$12x \equiv 27 \pmod{18}$: $d = 6 \nmid 27$, no solutions. $15x \equiv 27 \pmod{18}$: $d = 3$; $5x \equiv 9 \pmod 6$ gives $x \equiv 3 \pmod 6$, so $x \equiv 3, 9, 15 \pmod{18}$.

4.4 The field of quotients of an integral domain (§21)

Motivation. $\mathbb{Q}$ is built from $\mathbb{Z}$ by formal fractions. The same construction turns any integral domain D into the smallest field containing it; the lemmas below verify that the obvious definitions of $+$ and $\cdot$ on fractions are well defined.

Let D be an integral domain and $S = \{(a, b) \in D \times D : b \neq 0\}$; set $(a, b) \sim (c, d)$ iff $ad = bc$.

Lemma 4.17 (21.2)

$\sim$ is an equivalence relation on S.

Proof. Reflexive and symmetric by commutativity. If $ad = bc$ and $cf = de$, then $adf = bcf = bde$, so $d(af - be) = 0$; as $d \neq 0$, $af = be$. ∎

Let $F = S/\sim$ with classes $[(a, b)]$.

Lemma 4.18 (21.3)

$[(a, b)] + [(c, d)] = [(ad + bc, bd)]$ and $[(a, b)][(c, d)] = [(ac, bd)]$ are well-defined operations on F.

Proof. $bd \neq 0$. If $(a, b) \sim (a', b')$ and $(c, d) \sim (c', d')$, i.e. $ab' = a'b$, $cd' = c'd$, then $(ad + bc)b'd' = ab'dd' + bb'cd' = a'bdd' + bb'c'd = (a'd' + b'c')bd$, and $acb'd' = a'c'bd$. ∎

One checks that F is a field: $[(0, 1)]$ and $[(1, 1)]$ are the identities, $[(-a, b)]$ the additive inverse, and $[(b, a)]$ the inverse of $[(a, b)]$ for $a \neq 0$; the axioms reduce to identities in D.

Lemma 4.19 (21.4)

$i : D \to F$, $i(a) = [(a, 1)]$, is an isomorphism of D onto a subring of F.

Proof. $i(a) + i(b) = [(a + b, 1)]$, $i(a)i(b) = [(ab, 1)]$, and $[(a, 1)] = [(b, 1)]$ iff $a = b$. ∎

Theorem 4.20 (21.5)

Every integral domain D embeds in a field F in which every element is a quotient of two elements of D (a **field of quotients**).

Proof. Lemmas 4.17–4.19, with $[(a, b)] = i(a)i(b)^{-1}$. ∎

Theorem 4.21 (21.6)

If F is a field of quotients of D and L is a field containing D, then $\psi(a/b) = ab^{-1}$ (computed in L) gives an isomorphism of F onto a subfield of L with $\psi(a) = a$ for $a \in D$.

Proof. Well defined: $a/b = c/d$ means $ad = bc$, so $ab^{-1} = cd^{-1}$ in L. ψ preserves sums and products ($ab^{-1} + cd^{-1} = (ad + bc)(bd)^{-1}$, $ab^{-1}cd^{-1} = ac(bd)^{-1}$), and it is one-to-one because a field

homomorphism with $\psi(1) = 1$ has trivial kernel ($\psi(x) = 0$, $x \neq 0$, would give $1 = \psi(x)\psi(x^{-1}) = 0$). ■

Corollary 4.22 (21.8)

Every field containing an integral domain D contains a field of quotients of D.

Corollary 4.23 (21.9)

Any two fields of quotients of D are isomorphic (by an isomorphism fixing D).

Proof. If F, F' are fields of quotients, $\psi : F \to F'$ of Theorem 4.21 has image containing all ab^{-1}, i.e. all of F'. ■

4.5 Rings of polynomials (§22)

Motivation. To study equations $f(x) = 0$ algebraically we treat polynomials as formal objects and then "plug in" values. The evaluation homomorphism links the two views, so that finding a root of f becomes a question about the kernel of a ring homomorphism.

A **polynomial** with coefficients in R is $f(x) = \sum_{i \geq 0} a_i x^i$ with finitely many $a_i \neq 0$; polynomials are added and multiplied as usual, $(\sum a_i x^i)(\sum b_j x^j) = \sum_n (\sum_{i+j=n} a_i b_j) x^n$.

Theorem 4.24 (22.2)

$R[x]$ is a ring; it is commutative if R is, and has unity 1 if R does.

Proof. $\langle R[x], + \rangle$ is an abelian group (componentwise). Associativity: the coefficient of x^n in $(fg)h$ and in $f(gh)$ is $\sum_{i+j+k=n} a_i b_j c_k$. Distributivity follows from that of R. The rest is immediate. ■

$R[x_1, \ldots, x_n] = R[x_1, \ldots, x_{n-1}][x_n]$. If D is an integral domain, so is $D[x]$: the leading coefficient of fg is the product of the leading coefficients, and $\deg(fg) = \deg f + \deg g$.

Theorem 4.25 (22.4, evaluation homomorphism)

Let $F \leq E$ be fields and $\alpha \in E$. The map $\phi_\alpha : F[x] \to E$, $\phi_\alpha(\sum a_i x^i) = \sum a_i \alpha^i$, is a ring homomorphism with $\phi_\alpha(x) = \alpha$ and $\phi_\alpha(a) = a$ for $a \in F$.

Proof. Additivity is clear. For products, $\phi_\alpha(fg) = \sum_n (\sum_{i+j=n} a_i b_j) \alpha^n = (\sum a_i \alpha^i)(\sum b_j \alpha^j)$ since E is commutative. ■

α is a **zero** of f if $\phi_\alpha(f) = f(\alpha) = 0$.

Theorem 4.26 (22.11)

$x^2 - 2$ has no zero in $\mathbb{Q}$; thus $\sqrt{2}$ is irrational.

Proof. If $(m/n)^2 = 2$ in lowest terms, then $m^2 = 2n^2$, so m is even, $m = 2k$, $2k^2 = n^2$, and n is even, a contradiction. ■

4.6 Factorization of polynomials over a field (§23)

Motivation. Over a field, polynomials can be divided with remainder exactly like integers. This gives the factor theorem (roots correspond to linear factors) and makes irreducible polynomials the analogue of primes. Gauss's lemma and Eisenstein's criterion are the practical tools for proving irreducibility, which is what Chapter 6 will need to build field extensions.

Theorem 4.27 (23.1, division algorithm for $F[x]$)

Let $f, g \in F[x]$ with $g \neq 0$. There are unique $q, r \in F[x]$ with $f = gq + r$ and $r = 0$ or $\deg r < \deg g$.

Proof. Existence: the set $\{f - gs : s \in F[x]\}$ has an element r of least degree (or 0). If $\deg r = t \geq m = \deg g$, with leading coefficients c_t and b_m, then $r - (c_t/b_m)x^{t-m}g$ has smaller degree, a contradiction.

Uniqueness: if $gq + r = gq' + r'$, then $g(q - q') = r' - r$ has degree $< \deg g$ unless $q = q'$. ■

Corollary 4.28 (23.3, factor theorem)

$a \in F$ is a zero of f iff $x - a$ divides f.

Proof. $f = (x - a)q + r$ with $r \in F$; evaluating at a, $f(a) = r$. ■

Corollary 4.29 (23.5)

A nonzero $f \in F[x]$ of degree n has at most n zeros in F.

Proof. Induction: if a is a zero, $f = (x - a)q$ with $\deg q = n - 1$, and any other zero b satisfies $(b - a)q(b) = 0$, so $q(b) = 0$. ■

Corollary 4.30 (23.6)

A finite subgroup G of the multiplicative group F^* of a field is cyclic. In particular the multiplicative group of a finite field is cyclic.

Proof. By Theorem 2.17, $G \cong \mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_r}$ with prime powers d_i. Let $m = \operatorname{lcm}(d_i)$; then $a^m = 1$ for all $a \in G$, so $x^m - 1$ has $|G|$ zeros and $|G| \leq m$ (Corollary 4.29). But $m \leq \prod d_i = |G|$, with equality iff the d_i are pairwise relatively prime, in which case G is cyclic (Corollary 2.15). ■

A nonconstant $f \in F[x]$ is **irreducible over** F if it is not a product of two polynomials of lower degree in $F[x]$.

Theorem 4.31 (23.10)

A polynomial of degree 2 or 3 is reducible over F iff it has a zero in F.

Proof. A factorization into lower degrees must contain a linear factor $ax + b$, giving the zero $-b/a$; conversely use the factor theorem. ■

Theorem 4.32 (23.11, Gauss's lemma)

If $f \in \mathbb{Z}[x]$ factors in $\mathbb{Q}[x]$ into polynomials of degrees $r, s < \deg f$, it factors in $\mathbb{Z}[x]$ into polynomials of the same degrees.

Proof. Call $h \in \mathbb{Z}[x]$ primitive if its coefficients have gcd 1. The product of primitive polynomials is primitive: if a prime p divided all coefficients of gh, then reducing mod p gives $\bar{g}\bar{h} = 0$ in the integral domain $\mathbb{Z}_p[x]$, so $\bar{g} = 0$ or $\bar{h} = 0$. Now if $f = gh$ in $\mathbb{Q}[x]$, write $g = \frac{a}{b} g_1$, $h = \frac{c}{d} h_1$ with g_1, h_1 primitive in $\mathbb{Z}[x]$; then $bdf = ac\, g_1 h_1$. The gcd of the coefficients of the left side is a multiple of bd, and that of the right side is ac (as $g_1 h_1$ is primitive), so $bd \mid ac$ and $f = \frac{ac}{bd} g_1 \cdot h_1$ is a factorization in $\mathbb{Z}[x]$. ■

Corollary 4.33 (23.12)

If $f = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$ with $a_0 \neq 0$ has a zero in $\mathbb{Q}$, it has a zero $m \in \mathbb{Z}$, and $m \mid a_0$.

Proof. A rational zero gives a linear factor in $\mathbb{Q}[x]$, hence by Theorem 4.32 a factorization $(x - m)(x^{n-1} + \cdots - a_0/m)$ in $\mathbb{Z}[x]$ (the leading coefficients multiply to 1), so $m \in \mathbb{Z}$ and $m \mid a_0$. ■

Theorem 4.34 (23.15, Eisenstein criterion)

Let p be prime and $f = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ with $a_n \not\equiv 0$, $a_i \equiv 0 \pmod{p}$ for $i < n$, and $a_0 \not\equiv 0 \pmod{p^2}$. Then f is irreducible over $\mathbb{Q}$.

Proof. Otherwise $f = (b_r x^r + \cdots + b_0)(c_s x^s + \cdots + c_0)$ in $\mathbb{Z}[x]$ with $r, s < n$. Since $p^2 \nmid a_0 = b_0 c_0$ but $p \mid a_0$, exactly one of b_0, c_0, say b_0, is divisible by p. As $p \nmid a_n = b_r c_s$, $p \nmid b_r$; let i be least with $p \nmid b_i$, $0 < i \leq r < n$. Then $a_i = b_i c_0 + b_{i-1} c_1 + \ldots$, where p divides every term except $b_i c_0$; so $p \nmid a_i$, a contradiction. ■

Corollary 4.35 (23.17)

The cyclotomic polynomial $\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + \cdots + x + 1$ is irreducible over $\mathbb{Q}$ for every prime p.

Proof. $\Phi_p(x+1) = \frac{(x+1)^p - 1}{x} = x^{p-1} + \binom{p}{1} x^{p-2} + \cdots + \binom{p}{p-1}$, and $p \mid \binom{p}{k}$ for $0 < k < p$ while $\binom{p}{p-1} = p$; Eisenstein applies. A factorization of $\Phi_p(x)$ would give one of $\Phi_p(x+1)$. ■

Theorem 4.36 (23.18)

If $p(x)$ is irreducible in $F[x]$ and $p \mid rs$, then $p \mid r$ or $p \mid s$.

Proof. The set $\{ap + br\}$ is an ideal of $F[x]$, hence equals $\langle d \rangle$ (every ideal of $F[x]$ is principal, by the division algorithm as in Theorem 5.18). d divides p, so d is a unit or an associate of p. If $p \nmid r$, then d is a unit, say $d = 1 = ap + br$; then $s = aps + brs$ is divisible by p. ■

Corollary 4.37 (23.19)

If p is irreducible and divides $r_1 \cdots r_n$, then p divides some r_i.

Theorem 4.38 (23.20)

Every nonconstant $f \in F[x]$ is a product of irreducible polynomials, unique up to order and nonzero constant factors.

Proof. Existence by induction on degree: f is irreducible or a product of lower-degree factors.

Uniqueness: if $p_1 \cdots p_r = q_1 \cdots q_s$, then p_1 divides some q_j (Corollary 4.37), and as q_j is irreducible, $q_j = up_1$ with $u \in F^*$; cancel and induct. ■

4.7 Noncommutative examples (§24)

Motivation. Not every interesting ring is commutative: endomorphism rings, group rings and the quaternions are basic examples. The section ends with Wedderburn's theorem, which shows that noncommutativity cannot appear in finite division rings.

An **endomorphism** of an abelian group A is a homomorphism $A \to A$; $(\phi + \psi)(a) = \phi(a) + \psi(a)$.

Theorem 4.39 (24.1)

The endomorphisms $\mathrm{End}(A)$ of an abelian group form a ring under addition and composition.

Proof. $\phi + \psi$ is an endomorphism because A is abelian; $\langle \mathrm{End}(A), + \rangle$ is an abelian group with zero map and $-\phi$. Composition is associative, and $\phi(\psi + \chi) = \phi\psi + \phi\chi$ (as ϕ is additive) and $(\psi + \chi)\phi = \psi\phi + \chi\phi$ (by definition). ■

For a group G and commutative ring R with unity, the **group ring** RG consists of finite formal sums $\sum a_g g$ with $(\sum a_g g)(\sum b_h h) = \sum_k (\sum_{gh=k} a_g b_h)k$.

Theorem 4.40 (24.4)

RG is a ring.

Proof. Addition is componentwise. Associativity of multiplication: the coefficient of k in both triple products is $\sum_{ghl=k} a_g b_h c_l$, by associativity in G. Distributivity follows from that in R. ■

The **quaternions** $\mathbb{H}$ are $a_1 + a_2 i + a_3 j + a_4 k$ ($a_i \in \mathbb{R}$) with $i^2 = j^2 = k^2 = -1$, $ij = k = -ji$, $jk = i = -kj$, $ki = j = -ik$.

Theorem 4.41 (24.9)

$\mathbb{H}$ is a strictly skew field.

Proof. The map $a+bi+cj+dk \mapsto \begin{pmatrix} a+bi & c+di \\ -c+di & a-bi \end{pmatrix}$ is an additive injection into $M_2(\mathbb{C})$ preserving the multiplication rules of $1, i, j, k$; so $\mathbb{H}$ is isomorphic to a subring of $M_2(\mathbb{C})$, and associativity and distributivity hold. For $q = a_1+a_2 i+a_3 j+a_4 k \neq 0$ and $\bar{q} = a_1 - a_2 i - a_3 j - a_4 k$, $q\bar{q} = \bar{q}q = \sum a_i^2 > 0$, so $q^{-1} = \bar{q}/\sum a_i^2$. $ij \neq ji$. ■

Theorem 4.42 (24.10, Wedderburn)

Every finite division ring is a field.

Proof [Supplement]. (Witt.) Let D be a finite division ring and $Z = \{z : zx = xz\ \forall x\}$ its center, a finite field of q elements. D is a vector space over Z, so $|D| = q^n$. For $x \in D$ the centralizer $C(x) = \{y : yx = xy\}$ is a division ring containing Z, so $|C(x)| = q^d$; as D is a (left) vector space over $C(x)$, q^n is a power of q^d, so $d \mid n$, and $d < n$ if $x \notin Z$. The class equation of D^* (conjugation action, Theorem 3.20) gives

$$q^n - 1 = (q-1) + \sum \frac{q^n - 1}{q^{d_x} - 1},$$

summed over representatives of the noncentral classes, $d_x \mid n$, $d_x < n$. Let $\Phi_n(x) = \prod(x - \zeta)$ over the primitive n-th roots of unity ζ; from $x^n - 1 = \prod_{e \mid n} \Phi_e(x)$ and induction, $\Phi_n \in \mathbb{Z}[x]$ is monic, and $\Phi_n(x)$ divides $x^n - 1$ and $(x^n - 1)/(x^d - 1)$ in $\mathbb{Z}[x]$ for $d \mid n$, $d < n$. Hence the integer $\Phi_n(q)$ divides $q - 1$. But for $n > 1$ each $|q - \zeta| > q - 1 \geq 1$, so $|\Phi_n(q)| > q - 1$. Therefore $n = 1$ and $D = Z$ is commutative. ■

4.8 Ordered rings and fields (§25)

Motivation. Why can $\mathbb{C}$ not be ordered compatibly with its arithmetic? Ordered rings formalize the sign rules of $\mathbb{R}$; squares must be positive, which already excludes $i^2 = -1$.

A ring R is **ordered** if it has a subset P (positive elements) closed under addition and multiplication such that for each a exactly one of $a \in P$, $a = 0$, $-a \in P$ holds.

Theorem 4.43 (25.3)

In an ordered ring, squares of nonzero elements are positive, the characteristic is 0, and there are no divisors of 0.

Proof. If $a \in P$ then $a^2 \in P$; if $-a \in P$ then $a^2 = (-a)^2 \in P$. If $ab = 0$ with $a, b \neq 0$, then $\pm a, \pm b \in P$ give $\pm ab \in P$, contradicting $ab = 0$. If R has unity, $1 = 1^2 \in P$, so $n \cdot 1 \in P$ is never 0; in general $n \cdot a^2 \in P$ for $a \neq 0$. ■

Corollary 4.44 (25.4)

$\mathbb{Z}$ embeds in every ordered ring with unity, with its natural ordering induced, and the natural ordering is the only ordering of $\mathbb{Z}$. $\mathbb{C}$ cannot be ordered.

Proof. $n \mapsto n \cdot 1$ is one-to-one since the characteristic is 0, and $n \cdot 1 \in P$ for $n > 0$. In $\mathbb{Z}$, $1 = 1^2$ is positive, so all positive integers are. In $\mathbb{C}$, $-1 = i^2$ would be positive as well as 1. ■

Theorem 4.45 (25.5)

Defining $a < b$ iff $b - a \in P$: (trichotomy) exactly one of $a < b$, $a = b$, $b < a$ holds; (transitivity) $a < b$, $b < c \Rightarrow a < c$; (additivity) $a < b \Rightarrow a + c < b + c$; (multiplicativity) $a < b$, $0 < c \Rightarrow ac < bc$ and $ca < cb$.

Proof. Apply the axioms to $b-a$; $c-a = (c-b)+(b-a)$; $(b+c)-(a+c) = b-a$; $bc-ac = (b-a)c \in P$. ■

Theorem 4.46 (25.10)

If R is ordered by P and $\phi : R \to R'$ is a ring isomorphism, then $P' = \phi[P]$ orders R', and $\phi(a) < \phi(b)$ iff $a < b$.

Proof. ϕ preserves sums, products and negatives and is bijective, so P' satisfies the axioms; $\phi(b) - \phi(a) = \phi(b-a) \in P'$ iff $b - a \in P$. ■

Theorem 4.47 (25.13)

Let D be an ordered integral domain with positive set P and F its field of quotients. Then $P' = \{a/b : ab \in P\}$ is well defined, orders F, and induces the order of D; it is the only such order on F.

Proof. If $a/b = c/d$ $(ad = bc)$, then $ab\,d^2 = (ad)(bd) = (bc)(bd) = cd\,b^2$; since $b^2, d^2 \in P$, $ab \in P$ iff $cd \in P$ (multiplicativity and trichotomy). Closure: $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$ with $(ad+bc)bd = ab\,d^2 + cd\,b^2 \in P$, and $\frac{a}{b}\frac{c}{d} = \frac{ac}{bd}$ with $acbd = (ab)(cd) \in P$. Trichotomy for a/b follows from that of ab. For $a \in D$, $a/1 \in P'$ iff $a \in P$.

Uniqueness: in any order of F extending that of D, $a/b = ab \cdot (1/b)^2$ is positive iff ab is. ■

So $\mathbb{Q}$ has only its natural ordering.